

Functional Annotation Report: **argC2** (UniProt P59308, locus PP_3633) in *Pseudomonas putida* KT2440

0. Gene/Protein Identity Verification (PASSED)

The target gene symbol **argC2** is **consistent** with the UniProt protein description, organism, and domain architecture — there is **no** identity ambiguity or same-symbol homonym problem:

Attribute	Target (UniProt)	Verified against databases	Match
Accession	P59308	UniProtKB P59308	✓
Protein	N-acetyl-γ-glutamyl-phosphate reductase 2 (AGPR 2 / NAGSA dehydrogenase 2), EC 1.2.1.38	UniProt FUNCTION + KEGG K00145 (EC 1.2.1.38)	✓
Gene / locus	argC2 / PP_3633	KEGG <code>ppu:PP_3633</code> ; GenBank AAN69233	✓
Organism	<i>P. putida</i> KT2440 (taxid 160488)	UniProt organism = taxid 160488	✓
Family	NAGSA dehydrogenase family, Type 2	UniProt SIMILARITY; InterPro IPR010136 (AGPR type-2)	✓

All literature and database evidence below refers to the AGPR (ArgC) enzyme family and to *P. putida* KT2440 specifically. **The gene is correctly identified.**

1. Summary (Answer to the Research Question)

argC2 (P59308, PP_3633) encodes N-acetyl-γ-glutamyl-phosphate reductase (AGPR; also called N-acetyl-glutamate-5-semialdehyde / NAGSA dehydrogenase), a cytoplasmic, NADPH-dependent oxidoreductase (EC 1.2.1.38). It catalyzes the third (and only reductive) step of de novo **L-arginine biosynthesis** via the acetylated-ornithine route: the reduction of the acyl-phosphate **N-acetyl-L-glutamyl-5-phosphate** to the aldehyde **N-acetyl-L-glutamate-5-semialdehyde**, consuming NADPH and releasing inorganic phosphate and NADP⁺ (Rhea:21588). Its substrate is specifically the **N-acetylated** glutamyl-phosphate intermediate, not free glutamate derivatives, because the acetyl "protecting group" is retained throughout the pathway and recycled by ornithine acetyltransferase (ArgJ).

A distinctive feature uncovered here: *P. putida* KT2440 is **unusual in carrying two AGPR paralogs**. Comparative genomics shows that the **conserved, single-copy housekeeping AGPR** of *Pseudomonas* corresponds to **argC1/PP_0432** (91.5% identical to the sole AGPR of *P. aeruginosa*), whereas **argC2 is a divergent, lineage-restricted "Type-2" second isozyme** (~52% identical to the housekeeping enzyme). argC2 retains every catalytic determinant and is therefore predicted to catalyze the same reaction, most plausibly as an **accessory/conditionally-expressed isozyme**.

The functional annotation of P59308 itself is **computational/homology-based** (UniProt protein-existence level 3, HAMAP-Rule MF_01110); the reaction assignment is strongly supported by direct biochemistry on orthologous ArgC enzymes and by conservation of the catalytic machinery.

2. Primary Function: the Catalyzed Reaction and Substrate Specificity

2.1 Reaction

AGPR catalyzes (physiological direction = reductive, right→left as written by UniProt):

N-acetyl-L-glutamyl-5-phosphate + NADPH + H⁺ → N-acetyl-L-glutamate-5-semialdehyde + phosphate + NADP⁺ (EC 1.2.1.38; Rhea:21588; GO:0003942 "N-acetyl-γ-glutamyl-phosphate reductase activity")

UniProt (HAMAP MF_01110) states the FUNCTION verbatim: "*Catalyzes the NADPH-dependent reduction of N-acetyl-5-glutamyl phosphate to yield N-acetyl-L-glutamate 5-semialdehyde.*" KEGG places PP_3633 under EC 1.2.1.38 (oxidoreductase acting on the aldehyde/oxo group of donors, with NAD⁺/NADP⁺ as acceptor).

Mechanistically this is an unusual "reverse" dehydrogenase step: it reductively dephosphorylates a **high-energy acyl-phosphate** to an aldehyde. The reaction is energetically costly and is described in the literature as "*the high-energy-consuming third step in the arginine synthesis pathway*" (Vargas-Lagunas et al., 2017,).

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2.2 Substrate specificity

- **Acyl-phosphate substrate:** N-acetyl-L-glutamyl-5-phosphate (the product of N-acetylglutamate kinase, ArgB).
- **Reductant/cofactor:** NADPH (the enzyme carries an NAD(P)-binding Rossmann-type fold; InterPro IPR036291; GO:0051287 "NAD binding").
- **The N-acetyl group is obligatory:** the entire pathway operates on N-acetylated intermediates so that the α-amino group of glutamate is masked, preventing spontaneous cyclization of the glutamate-5-semialdehyde to Δ¹-pyrroline-5-carboxylate (which would divert flux toward proline). ArgC therefore has strict specificity for the *acetylated* semialdehyde/acyl-phosphate pair, distinguishing it from the analogous non-acetylated enzymes of proline/ornithine metabolism.

2.3 Enzyme identity — experimental support from orthologs

Because P59308 has not itself been assayed, the reaction assignment rests on **direct biochemical characterization of orthologous ArgC enzymes**, all of which complement *E. coli argC* mutants and display AGPR/semialdehyde-dehydrogenase activity: - *Corynebacterium glutamicum* ArgC: "*the cloned gene indeed codes for N-acetylglutamate 5-semialdehyde dehydrogenase*" (Chun et al., 1998,

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- *Streptomyces clavuligerus* ArgC: "The *argC* gene ... encoding N-acetylglutamyl-phosphate reductase (AGPR) has been cloned by complementation of *argC* mutants" (Ludovice et al., 1992,

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- *Sinorhizobium meliloti* ArgC: "argC encodes N-acetyl-gamma-glutamyl phosphate reductase, the enzyme that catalyzes the ... third step in the arginine synthesis pathway" (Vargas-Lagunas et al., 2017,

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3. Structural Basis and Catalytic Mechanism (inference from sequence/structure)

P59308 is a **313-residue** soluble protein built on the semialdehyde-dehydrogenase fold (Pfam Semialdehyde_dh / Semialdehyde_dhC), comprising an N-terminal NAD(P)-binding (Rossmann-like) domain and a C-terminal catalytic/dimerization domain (InterPro IPR058924 "AGPR dimerisation domain"). AGPRs are typically **homodimers**, consistent with the dedicated dimerization domain.

Two lines of structure/evolution evidence support catalytic competence: 1. **Conserved active-site motif.** The diagnostic AGPR signature **P-G-C-Y-P-T** (InterPro IPR023013 "AGPR_AS") is present at residues 115–120 (context ...RVSN **PGCYPT** GAI...), placing the **invariant catalytic cysteine at position 117**. Ludovice et al. (1992) identified this PGCYPT motif as conserved across all AGPRs and proposed it as "the active center of the protein" (

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The active-site cysteine forms the covalent thioacyl/tetrahedral intermediate during hydride transfer from NADPH to the acyl-phosphate carbon. 2. **Cofactor-binding elements.** A C-terminal glycine-rich segment (K-G-A-S-G, residues 291–296) and the NAD(P)-binding domain superfamily fold (IPR036291) provide the dinucleotide-binding platform (GO:0051287).

Structural-model corroboration. The AlphaFold model (AF-P59308-F1, v6) is **very high confidence** — global pLDDT **96.1**, with 93% of residues at pLDDT ≥ 90 and the catalytic segment (residues 115–120, PGCYPT/Cys117) modeled at pLDDT 97–99. This indicates a

compact, well-ordered globular fold with a structurally well-defined active site, providing structure-based support — beyond sequence homology — that argC2 is a folded, catalytically configured AGPR.

Type-1 vs Type-2 distinction (a defining feature of argC2): the classical (Type-1) ArgC — e.g., *P. aeruginosa* PA0662 and *P. putida* argC1/PP_0432 — carries the canonical N-terminal Rossmann fingerprint **G-x-G-x-x-G** ("GIVGGTGY", residues 5–10). **argC2 lacks this canonical N-terminal motif**, the structural hallmark of the InterPro **Type-2** AGPR subfamily (IPR010136). Despite the rearranged dinucleotide-binding region, argC2 retains the catalytic Cys and C-terminal binding motifs, so it is predicted to be a functional, if divergent, NADPH-dependent AGPR.

4. Biological Process and Pathway Context

AGPR operates in the **de novo L-arginine biosynthetic pathway** (glutamate → ornithine → citrulline → arginine) via **N-acetylated intermediates**, the general route in microorganisms (Xu, Labedan & Glansdorff, 2007,

[P 17347518](#)

Caldovic & Tuchman, 2003,).

[P 12633501](#)

KEGG assigns PP_3633 to **module M00028** ("Ornithine biosynthesis, glutamate => ornithine") and pathway **ppu00220** (Arginine biosynthesis); UniProt places it at **step 3 of 4** in the sub-pathway "N(2)-acetyl-L-ornithine from L-glutamate."

The reaction sequence and the *P. putida* genes involved:

1. **Glutamate acetylation** → N-acetylglutamate (NAG). *P. putida* uses the **cyclic/economical** variant in which **ornithine acetyltransferase ArgJ (PP_1346)** transfers the acetyl group from N-acetylornithine back onto glutamate — "NAG can be produced by ... NAG synthase (NAGS) and ... ornithine acetyltransferase (OAT)" (Caldovic & Tuchman, 2003,

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2. **NAG phosphorylation** by N-acetylglutamate kinase **ArgB (PP_5289)** → N-acetyl-L-glutamyl-5-phosphate.

3. **Reduction by AGPR (argC / argC2) → N-acetyl-L-glutamate-5-semialdehyde** ← the step catalyzed by P59308.
4. **Transamination** by acetylornithine aminotransferase **ArgD (PP_0372)** → N-acetylornithine.
5. Conversion to **ornithine** (via ArgJ acetyl-recycling and/or ArgE deacetylation), then carbamoylation → citrulline → argininosuccinate → **L-arginine**.

Thus the precise metabolic role of argC2 is to supply **N-acetyl-glutamate-5-semialdehyde** — the amino-group acceptor for the downstream transaminase — as the **sole reductive (NADPH-consuming) node** of ornithine biosynthesis. This is a strictly anabolic, biosynthetic role (not catabolic and not regulatory).

5. Subcellular Localization

Cytoplasm. UniProt SUBCELLULAR LOCATION = Cytoplasm (GO:0005737, cytoplasm). This is expected and well supported: AGPR is a soluble metabolic dehydrogenase with no signal peptide, no transmembrane segments, and a cofactor-dependent globular fold; all substrates/products (acetyl-glutamyl-phosphate, semialdehyde, NADPH/NADP⁺, Pi) are cytosolic metabolites. The enzyme therefore carries out its function in the bacterial cytosol.

6. Paralogy, Evolution, and the Special Status of *argC2*

A central result of this investigation concerns **why *P. putida* has an "argC2" at all**:

- **Copy number is unusual.** Across representative genomes, *P. aeruginosa* (PA0662), *P. protegens* (PFL_5611), *P. stutzeri* (PSPTO_0604), *P. syringae* (P syr_4569) and *E. coli* (b3958 = argC) each encode **one** AGPR (K00145). **Only *P. putida* KT2440 encodes two** — PP_0432 (argC1) and PP_3633 (argC2).
- **argC1 is the conserved ortholog.** The single-copy *P. aeruginosa* AGPR (PA0662) is **91.5%** identical to argC1/PP_0432 but only **52.3%** identical to argC2/PP_3633 (Needleman–Wunsch, this work). argC1 and PA0662 share the Type-1 N-terminal Rossmann motif. Hence **argC1 is the housekeeping ArgC; argC2 is a divergent, taxonomically restricted second isozyme** (Type 2), sharing only ~53% identity with argC1.

- **Genomic context suggests an accessory role.** argC2/PP_3633 is **not** in an arg operon (the core arg genes are dispersed). Its immediate neighbors are a **divergent LysR-family transcriptional regulator (PP_3632)**, a **GNAT-family acetyltransferase (PP_3634)**, and a **sulfonate/nitrate NitT/TauT ABC transporter (PP_3635–37)**. Co-location with a dedicated regulator and an acetyltransferase (a potential source/recycler of acetyl groups) is consistent with a **conditionally regulated or sub-/neo-functionalized** copy, rather than the constitutive biosynthetic enzyme — though this is inference from gene neighborhood and awaits experimental test.

Interpretation. argC2 most likely arose by gene duplication/divergence (or acquisition) in the *P. putida* lineage and provides a **second, divergent AGPR activity**. Its retained catalytic machinery makes it a bona fide predicted AGPR (EC 1.2.1.38); its divergence and genomic setting hint at a specialized physiological trigger. Whether it acts on the canonical N-acetyl-glutamyl-phosphate substrate exclusively, or has shifted specificity, is not experimentally resolved.

7. Evidence Summary and Confidence

Claim	Evidence type	Strength
Enzyme = AGPR, EC 1.2.1.38, reduces N-acetyl-glutamyl-5-phosphate → N-acetylglutamate-5-semialdehyde (NADPH)	UniProt/HAMAP + KEGG annotation; direct biochemistry on orthologs (PMIDs 9818083, 1339424, 29121239)	High
Catalytic Cys117 in conserved PGCYPT motif; NAD(P)-binding fold	Sequence/structure inference; motif defined by P 1339424	High
Role = step 3 of acetylated arginine/ornithine pathway; substrate is N-acetylated	KEGG module M00028; pathway reviews (PMIDs 17347518, 12633501)	High
Localization = cytoplasm	UniProt/GO; sequence features (no signal/TM)	High
Well-ordered AGPR fold with defined active site	AlphaFold AF-P59308-F1, global pLDDT 96.1 (active site 97–99)	High (structural model)
argC2 is the divergent Type-2 second isozyme; argC1/PP_0432 is the conserved housekeeping ortholog	Comparative genomics + pairwise identity (this work)	High (bioinformatic)
Accessory/conditional role implied by genomic context	Gene-neighborhood inference	Moderate / hypothesis
Direct enzymology of P59308 itself; substrate-specificity kinetics; in vivo phenotype of a <i>P. putida</i> argC2 mutant	—	Not available (gap)

8. Supported and Refuted Hypotheses

- **Supported:** argC2 is an AGPR (EC 1.2.1.38) acting in arginine biosynthesis; it is cytoplasmic; it retains the catalytic Cys/PGCYPT machinery; it is a divergent Type-2 paralog distinct from the conserved housekeeping AGPR.

- **Supported (comparative-genomic):** the conserved single-copy *Pseudomonas* AGPR corresponds to argC1/PP_0432, identifying argC2 as the lineage-specific "extra" copy.
- **Not refuted, unresolved:** that argC2 is an accessory/conditionally-expressed isozyme (plausible from genomics; not experimentally tested). Whether argC2 can fully substitute for argC1 in arginine prototrophy is unknown.

9. Limitations and Future Directions

- P59308 is annotated by homology (existence level 3); **no direct enzymology, structure, or targeted mutant** of this specific protein exists in the retrieved literature.
- Attempts to retrieve high-throughput RB-TnSeq fitness data (LBL Fitness Browser) were blocked (HTTP 403); direct genetic-essentiality evidence for argC2 vs argC1 could not be obtained. The task prioritizes precise over high-throughput data, so this does not affect the core annotation.
- **Recommended experiments:** (i) recombinant expression + spectrophotometric AGPR assay (NADPH oxidation with N-acetyl-glutamyl-phosphate) and steady-state kinetics; (ii) single and double *argC1/argC2* deletions with arginine-auxotrophy complementation; (iii) reporter/RNA-seq to identify the condition(s) and the LysR regulator controlling argC2; (iv) crystallography/AlphaFold-guided analysis of the atypical Type-2 N-terminal dinucleotide-binding region.

Key References

- Chun et al. 1998, *Mol. Cells* — *C. glutamicum* argC encodes NAGSA dehydrogenase
([P 9818083](#)).
- Ludovice et al. 1992, *J. Bacteriol.* — *S. clavuligerus* argC = AGPR; conserved PGCYPT active-site motif
([P 1339424](#)).
- Vargas-Lagunas et al. 2017, *FEMS/Mol. genetics* — argC = N-acetyl-γ-glutamyl-phosphate reductase, energy-consuming third step
([P 29121239](#)).

- Xu, Labedan & Glansdorff 2007, *Microbiol. Mol. Biol. Rev.* — arginine biosynthesis via acetylated intermediates

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- Caldovic & Tuchman 2003, *Biochem. J.* — N-acetylglutamate and the acetylated arginine pathway; NAGS vs OAT

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- Database records: UniProt **P59308** (HAMAP MF_01110); KEGG **ppu:PP_3633** (K00145, EC 1.2.1.38, module M00028); InterPro IPR010136/IPR023013/IPR058924/IPR036291.

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